Name: Abdul Hakeem Faheem

Roll NO: 039

Class: BSSE (2A)

Subject: Programing Fundamental

Assignment no 1:

Task 1:

```
#include <iostream>
using namespace std;
int main() {
   int x, y;
   cout<<"enter value of y: ";
   cin >> y;
   if (y == 10)
      x = 0;
   else
      x = 1;
   cout << "x = " << x;
   return 0;
}
```

Task 2:

```
#include <iostream>
using namespace std;

int main() {
   int temperature;
   cout<<"enter temprature: ";
   cin >> temperature;
   if (temperature >= -50 && temperature <= 150)
```

```
    cout << "The number is valid";
        else
        cout<<"invalid number";
  return 0;
}
```

Task 3:

```
#include <iostream>
using namespace std;

int main() {
  int a, b;
  cout<<"enter the value of a and b: ";
  cin >> a >> b;
  if (a > b)
    cout << "Larger: " << a <<endl<< "Smaller: " << b;
  else
    cout << "Larger: " << b <<endl<< "Smaller: " << a;
  return 0;
}
```

Task 4:

```
#include <iostream>
using namespace std;
int main()
{
  double a = 10, b = 10;
  a = a + 0.2;
  if (a > b)
    cout << "a is larger";
  else
```

```
        cout << "b is larger";
    return 0;
}
```

Task 5:

```
#include <iostream>
using namespace std;

int main() {
    int num1, num2;
    cout<<"enter numbers: ";
    cin >> num1 >> num2;
    if (num2 != 0)
        cout << "Result: " << (float)num1 / num2;
    else
        cout << "Division by zero is not possible.";
    return 0;
}
```

Task 6:

```
#include <iostream>
using namespace std;
int main() {
    int a = 2, b = 4, c = 6;
    if (a == 4 || b > 2)
        cout << "A) True" << endl;
    else
        cout << "A) False" << endl;
    if (6 <= c && a != 3)
        cout << "B) True" << endl;
    else
```

```
        cout << "B) False" << endl;
    if (1 != b && c != 3)
        cout << "C) True" << endl;
    else
        cout << "C) False" << endl;
    if (a >= -1 || a <= b)
        cout << "D) True" << endl;
    else
        cout << "D) False" << endl;
    if (!(a > 2))
        cout << "E) True" << endl;
    else
        cout << "E) False" << endl;
    return 0;
}
```

Task 7:

```
#include <iostream>
using namespace std;
int main() {
    int r1, r2, r3;
    cout<<"enter r1, r2, r3: ";
    cin >> r1 >> r2 >> r3;
    if (r1 < r2 && r1 < r3)
        cout << "1st: Runner1"<<endl;
    else if (r2 < r1 && r2 < r3)
        cout << "1st: Runner2"<<endl;
    else
        cout << "1st: Runner3"<<endl;
    return 0;
```

```
}
```

Task 8:

```
#include <iostream>
using namespace std;
int main() {
   int choice;
   cout << "1.Circle\n2.Rectangle\n3.Triangle\n";
   cin >> choice;
   if (choice == 1) {
      double r;
      cout<<"you chose circle"<<endl;
      cout<<"enter value of r: ";
      cin >> r;
      cout << 3.14159 * r * r;
   }
   else if (choice == 2) {
      double l, w;
      cout<<"you chose Rectangle"<<endl;
      cout<<"enter length and width: ";
      cin >> l >> w;
      cout << l * w;
   }
   else if (choice == 3) {
      double b, h;
      cout<<"you chose Triangle"<<endl;
      cout<<"enter base and heigt: ";
      cin >> b >> h;
      cout << 0.5 * b * h;
   }
```

```
    return 0;
}
```

Task 9:

```
#include <iostream>
using namespace std;
int main()
{
    for(int  ch = 'A'; ch <= 'Z'; ch++) {
        cout << (char)ch << " = " << ch << endl;
    }
    return 0;
}
```

Task 10:

```
#include <iostream>
using namespace std;
int main()
{
    double mass, weight;
    cout<<"Enter Mass: ";
    cin >> mass;
    weight = mass * 9.8;
    cout << "Weight: "<< weight << endl;
    if (weight > 1000)
        cout << "Too heavy";
    else if (weight < 10)
        cout << "Too light";
    return 0;
}
```

Task 11:

```
#include <iostream>
using namespace std;
int main() {
   double sales;
   cout<<"Enter Sales: ";
   cin >> sales;
   if (sales > 50000)
      sales = sales + (0.25 * sales) + 250;
   cout << sales;
   return 0;
}
```

Task 12:

```
#include <iostream>
using namespace std;
int main() {
   double income;
   int years;
   cout<<"enter income and years: ";
   cin >> income >> years;
   if (income >= 35000 && years > 5)
      cout << "Qualified";
   else
      cout << "Not Qualified";
   return 0;
}
```

Task 13:

```
#include <iostream>
using namespace std;
int main() {
   int model;
   cout<<"enter Model number: ";
   cin >> model;
   if (model == 300)
      cout << "Picture-in-picture, Stereo sound, Remote control";
   else if (model == 200)
      cout << "Stereo sound, Remote control";
   else if (model == 100)
      cout << "Remote control only";
   else
         cout<<"enter valid number";
   return 0;
}
```

Task 14:

```
#include <iostream>
using namespace std;
int main() {
   int m, d, y;
   cout<<"enter month, day, and the last 2 digit of year ";
   cin >> m >> d >> y;
   if (m * d == y)
      cout << "MAGIC";
   else
      cout << "NOT MAGIC";
```

```
    return 0;
}
```

Task 15:

```
#include <iostream>
using namespace std;
int main() {
    double sales, commission;
    cout<<"enter sales: ";
    cin >> sales;
    if (sales <= 10000)
        commission = 0.10;
    else if (sales <= 15000)
        commission = 0.15;
    else
        commission = 0.20;
    cout << commission;
    return 0;
}
```

Task 16:

```
#include <iostream>
using namespace std;

int main() {
    double balance, fee = 10;
    int checks;
    cout<<"enter balane and checks: ";
    cin >> balance >> checks;
    if (checks < 20)
        fee += checks * 0.10;
```

```
    else if (checks <= 39)
        fee += checks * 0.08;
    else if (checks <= 59)
        fee += checks * 0.06;
    else
        fee += checks * 0.04;
    if (balance < 400)
        fee += 15;
    cout << "Fees: " << fee;
    return 0;
}
```

Task 17:

```
#include <iostream>
using namespace std;
int main() {
    int choice;
    cout<<"Enter choice: ";
    cin >> choice;
    if (choice == 1)
        cout << "249.00";
    else if (choice == 2)
        cout << "199.00";
    else if (choice == 3)
        cout << "49.00";
    return 0;
}
```

Task 18:

```
#include <iostream>
using namespace std;
```

```
int main()
{
   int choice, months;
   cout<<"enter choice: ";
   cin >> choice;
   if (choice == 4)
   cout<<"enter month: ";
   cin >> months;
   if (choice == 1)
      cout << months * 40;
   else if (choice == 2)
      cout << months * 20;
   else if (choice == 3)
      cout << months * 30;
   return 0;
}
```

Task 19:

```
#include <iostream>
using namespace std;
int main() {
   int pkg;
   double gb, total;
   cout<<"enter pkg and gb: ";
   cin >> pkg >> gb;
   if (pkg == 1) {
      total = 39.99;
      if (gb > 4)
         total += (gb - 4) * 10;
   }
```

```
    else if (pkg == 2) {
        total = 59.99;
        if (gb > 8)
            total += (gb - 8) * 5;
    }
    else if (pkg == 3) {
        total = 69.99;
    }
    cout << total;
    return 0;
}
```

Task 20:

```
#include <iostream>
using namespace std;
int main()
{
    int sec;
    cout<<"enter seconds: ";
    cin >> sec;
    if (sec >= 86400)
        cout << "Days: " << sec / 86400 << endl;
    if (sec >= 3600)
        cout << "Hours: " << sec / 3600 << endl;
    if (sec >= 60)
        cout << "Minutes: " << sec / 60 << endl;
    return 0;
}
```